

INTRODUCTION TO DATA SCIENCE AND MACHINE LEARNING

Project Presentation

Group 1

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Self created variables

Seasonality (Fourier Features)

- $\sin_{1y}$ / $\cos_{1y}$ – Smooth annual demand cycle
- $\sin_{2y}$ / $\cos_{2y}$ – Sharper seasonal peaks

Structural Differences

- WG (Product group dummies)

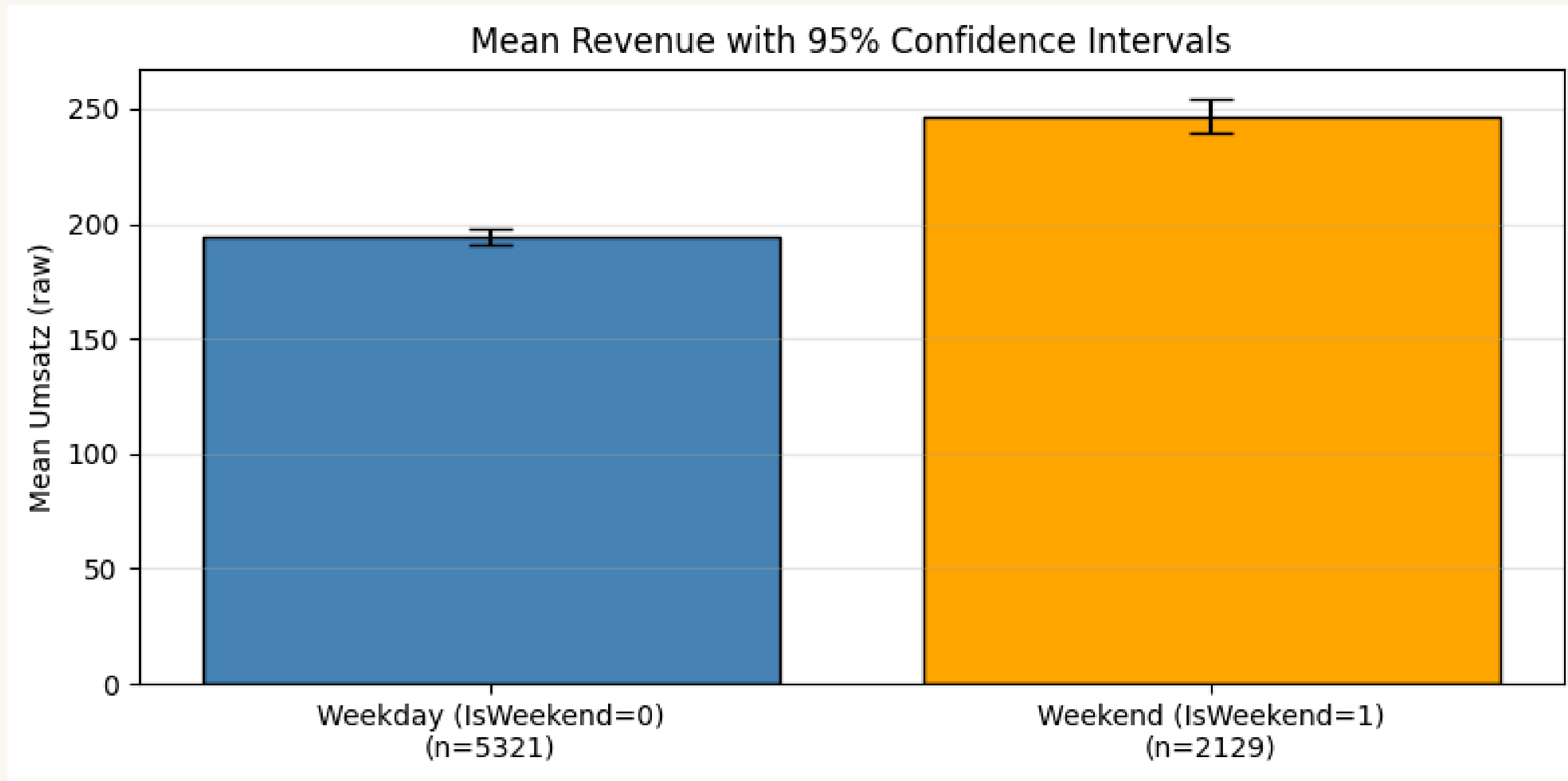
Time Dependence (Lags)

- Revenue_lag1 – Yesterday's demand (short-term persistence)
- Revenue_lag7 – Same weekday last week (weekly routines)

Calendar and Event Features

- IsWeekend – Weekend vs. weekday behavior
- IsNewYears – Special demand on Dec 31
- IsHalloween – Short-term seasonal event (Oct 24–31)
- KielerWoche – Local festival impact
- holiday
- Easter

Two self created variables



Linear model optimization

Model equation:

$$\begin{aligned} \text{Umsatz}_t = & \beta_0 + \beta_1 \cdot \text{holiday}_t + \beta_2 \cdot \text{IsWeekend}_t + \beta_3 \cdot \text{IsNewYears}_t \\ & + \beta_4 \cdot \text{Easter}_t + \beta_5 \cdot \text{IsHalloween}_t + \beta_6 \cdot \text{KielerWoche}_t \\ & + \beta_7 \cdot \sin_{1y,t} + \beta_8 \cdot \cos_{1y,t} + \beta_9 \cdot \sin_{2y,t} + \beta_{10} \cdot \cos_{2y,t} \\ & + \beta_{11} \cdot \text{Revenue_lag1}_t + \beta_{12} \cdot \text{Revenue_lag7}_t \\ & + \sum_{g=2}^G \gamma_g \cdot \text{WG}_g + \varepsilon_t \end{aligned}$$

Adjusted R²: 0.834

Handling missing values

Code

```
predictors = [  
    "holiday",  
    "IsWeekend",  
    "IsNewYears",  
    "Easter",  
    "KielerWoche",  
    "sin_1y",  
    "cos_1y",  
    "sin_2y",  
    "Revenue_lag1",  
    "Revenue_lag7"]
```

NaN values for each variable

NaN values in each column

Column	NaN Count
Datum	0
Umsatz	0
KielerWoche	0
holiday	0
Easter	0
IsWeekend	0
IsNewYears	0
IsHalloween	0
sin_1y	0
cos_1y	0
sin_2y	0
Revenue_lag1	6
Revenue_lag7	42

Neural Network Optimization

Source code defining the neural network

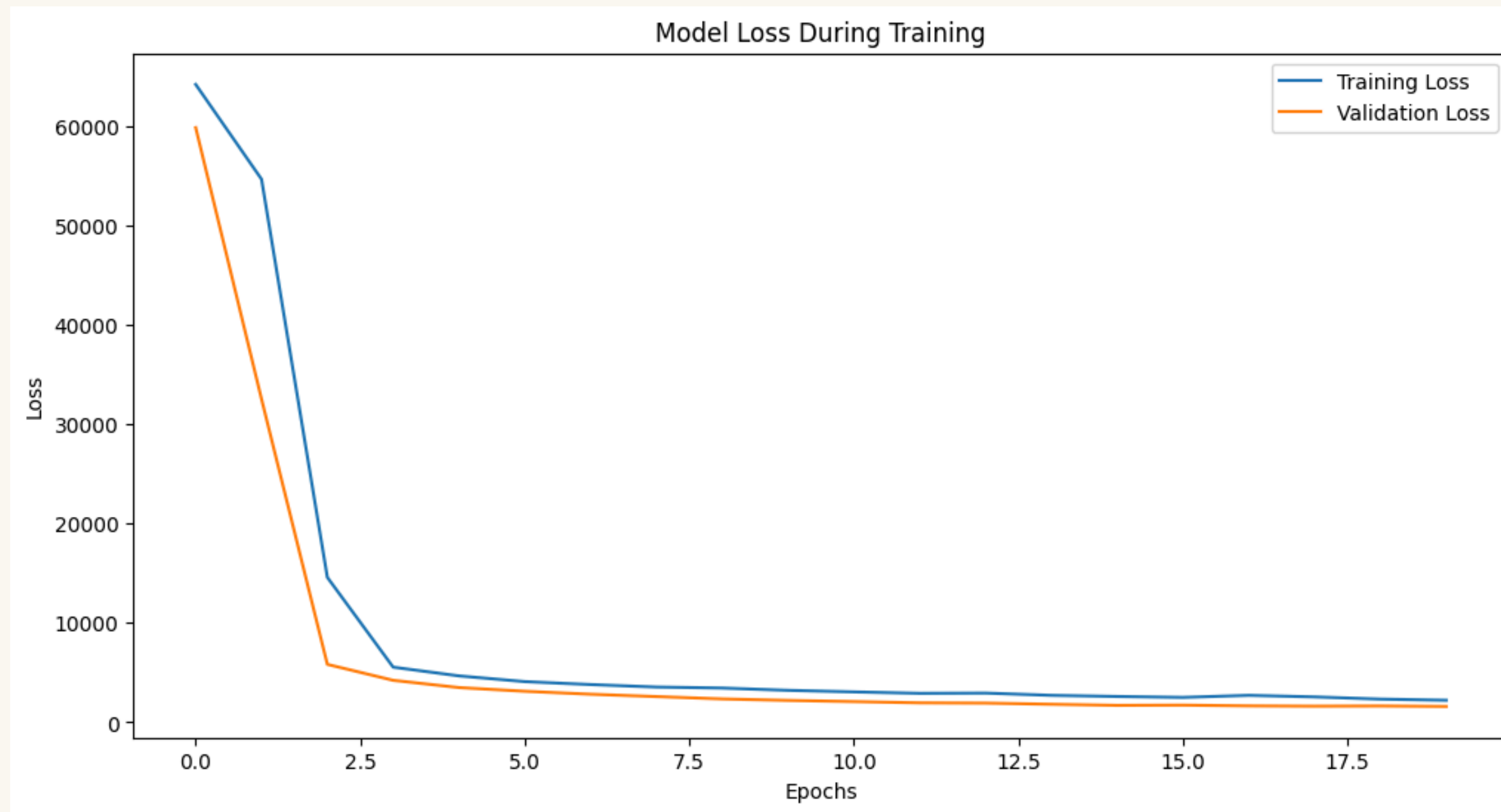
```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import InputLayer, Dense, BatchNormalization
from tensorflow.keras.optimizers import Adam

model = Sequential([
    InputLayer(shape=(training_features.shape[1], )),
    BatchNormalization(),
    Dense(10, activation='relu'),
    Dense(4, activation='relu'),
    Dense(1)
])

model.summary()
```

Neural Network Optimization

Loss function plots for training and validation sets



Neural Network Optimization

MAPE scores for the overall validation set and each product group

MAPE on the Validation Data: 24.60%		
MAPE per product category (Warengruppe)		
	Warengruppe	MAPE_Validation
0	1	40.634784
1	2	13.681085
2	3	16.223104
3	4	33.066130
4	5	14.525836
5	6	45.641443

Warengruppe	Avg_Daily_Revenue
1	122.142509
2	402.497911
3	163.340920
4	87.954042
5	276.795509
6	66.916635

Worst Fails

Clean and simple data management

Especially when working in a group, naming of dataframes can easily get out of hand. A clear structure can avoid hours of extra work.

Best Improvements

Understand your data

Take your time to understand the data. Realizing which variables explain the revenue based on graphs and descriptive statistics brought the biggest improvement.

Do you have
any questions?

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